A simulation-based hydrogen refuelling station digital twin integrating process dynamics, consequence screening and SAGA safety guidance

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Working manuscript. This manuscript reports development and bounded evaluation of a simulation-based safety digital twin. Facility-specific ethics and expert review are deployment prerequisites, not measured endpoints here; no human-effectiveness claim is made.

Abstract

A simulation-based hydrogen refuelling station digital twin was developed to connect process dynamics, hazard identification, consequence screening and staged safety guidance through SAGA, a language-model assistant. The framework distinguishes observed signals, inferred scenarios, calculated consequences, requested actions and verified simulated outcomes. Evaluation combined public experiments, confidential station-side holdouts and public incident records. Across 12 held-out Type-IV fills, pressure, temperature and state-of-charge RMSE were 3.841 MPa, 4.833 °C and 3.850 percentage points. A same-site recharge forecast achieved 0.184 MPa MAE on 394 holdout cases. Consequence evaluation identified both bounded successes and failed transfers. On 34 historical incidents, the SAGA response-structure proxy was 0.760, compared with 0.401 for alarm-only text. The contribution is an executable, traceable safety-analysis workflow; complete station control, site-specific distances and operator effectiveness remain unvalidated.

Keywords: hydrogen refuelling station; digital twin; HAZOP; consequence analysis; language model; safety decision support

1 Introduction

Hydrogen refuelling stations combine high-pressure storage and compression with rapid changes in flow, pressure and temperature during vehicle filling. A disturbance can involve several parts of this network: a storage leak changes inventory and available pressure, isolation changes the connected source, and residual hydrogen can remain after an

emergency-stop command. Safety analysis must therefore relate an abnormal indication to the physical process, the possible consequences and the actions available to an operator.

The station decomposition in ISO 19880-1 provides a useful engineering boundary, while SAE J2601 addresses the vehicle-fuelling protocol [1, 2]. Thermodynamic modelling has established approaches to station-to-vehicle transfer, cascade optimisation, vehicle-side pressure losses and tank heat transfer [3, 4, 5, 6]. Consequence packages such as HyRAM+ provide models of unintended releases and their thermal and pressure effects [7, 8]. These capabilities address different questions. Predicting tank temperature does not establish the accuracy of an emergency response, and calculating a free-jet consequence does not establish a safe distance for a particular facility.

The present work addresses the connection between these capabilities. A conventional alarm identifies a threshold crossing, but leaves the operator to relate it to concurrent signals, possible causes, equipment isolation and restart requirements. A language-model explanation can organise this information, yet it introduces another failure mode if its statements are disconnected from the current process or if it invents a numerical consequence. An integrated system needs an explicit account of who owns each state and how every recommendation can be checked.

This research developed a virtual H70 station coupled to the Safety Analysis and Guidance Assistant (SAGA). The system links a conserved process model to virtual instruments, HAZOP rules, consequence calculations and staged response plans. An operator can then execute a virtual action and inspect its physical and logical consequences. The research question is whether these links can be implemented reproducibly and supported by clearly bounded experimental and incident evidence. The objective is broader than a three-dimensional visualisation and narrower than a field-qualified automatic safety system.

Three contributions organise the study. First, a shared process state connects normal operation, fault injection, detection, isolation and consequence recalculation. Second, HAZOP deviations and incident evidence are converted into structured response plans with explicit execution and completion conditions; SAGA explains this context through separate station and sensor interfaces. Third, a multi-source evaluation distinguishes numerical component performance, station-side transfer, software correctness and response structure. Negative results are retained to define the limits of each contribution.

The work uses a simulation-based digital twin: every live sensor and actuator in the demonstrator is virtual. Confidential operational records are used for offline calibration and holdout assessment, not as a live connection to a station. Synchronized station-to-vehicle raw data were not available for a complete final-system holdout. Accordingly, the paper evaluates components and integration boundaries separately and makes no claim of SAE-certified control, validated site separation distances or improved human emergency performance.

2 Research design and system architecture

2.1 System boundary and information ownership

The reference topology consists of finite tube-trailer supply, a three-stage compressor, low-, medium- and high-pressure banks, a finite common header, pre-cooling and two dispenser–hose–vehicle circuits. The circuits retain independent filling requests and dispatch state. A finite compressor-package inventory represents gas trapped between isolation boundaries.

Storage volumes, equipment parameters and the visual layout are declared model inputs rather than an as-built reconstruction.

The Python process service owns the mass, energy, pressure, temperature and flow states. Virtual sensors expose these states with identity, units, simulation time, origin and quality. The HAZOP layer evaluates deviations on those signals. The consequence layer receives a release snapshot; SAGA receives the resulting process, hazard and consequence evidence. Browser views display the same job frames and do not independently calculate process dynamics. Table 1 identifies the responsibility at each interface; Fig. 1 shows the information and virtual-action paths.

Table 1: Responsibilities and checkable outputs in the integrated workflow.

Layer	Main responsibility	Checkable output
Process model	Inventory and energy evolution	State trajectories, signed flows and balance residuals
Virtual instrumentation	Sample and qualify process evidence	Value, unit, time, origin and quality
HAZOP and response logic	Detect deviations and select staged plans	Rule state, supporting signals and completion conditions
Consequence analysis	Evaluate a declared release boundary	Calculation status, model inputs, sampled endpoints and uncertainty
SAGA	Explain the question using supplied evidence	Grounded answer and claim-limit checks
Virtual response	Execute and assess requested actions	Command, motion, feedback, flow confirmation and replay

Three interfaces are kept distinct: the station-level conversation, analysis of a selected sensor and the standalone SAGA document-search service. This separation prevents a periodic station report from replacing a user's question or a general document answer from being presented as current sensor analysis. The direct digital-twin interfaces use supplied evidence without invoking the standalone document-search and review sequence on every exchange.

2.2 Experimental strategy

The evaluation was organised by observable boundary rather than by a single system-wide accuracy score. Public filling experiments test vehicle-tank thermodynamics with measured inlet conditions. Restricted operational traces test station-side behaviour using disjoint time-ordered records. Public release experiments test depletion and consequence components. Historical incident records test response mapping and a deterministic language-output proxy. Software regressions test whether these interfaces preserve their contracts.

This design separates four evidence roles. Calibration records estimate global parameters; frozen validation records evaluate the resulting model; post-outcome diagnostics explain a failure; and contextual sources motivate scenarios without supplying a numerical score. A source cannot change role merely because a later model fits it better. Protocols record eligibility, endpoints, thresholds, input assumptions and code/data hashes where an outcome-unseen evaluation was possible. Previously accessed records are explicitly classified as diagnostic or developmental.

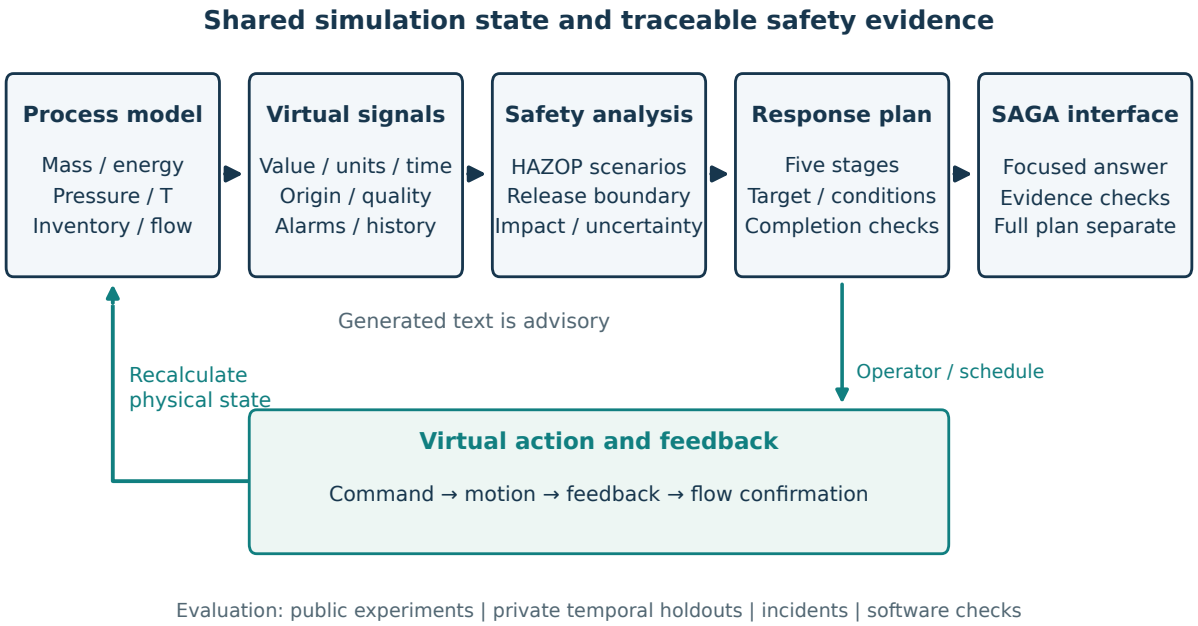


Figure 1: Shared-state workflow. Process dynamics and structured response logic retain state ownership; consequence results and SAGA explanations carry the scope of their evidence. Operator-issued or prescheduled virtual commands enter the action-and-feedback loop shown here.

3 Coupled process and virtual safety model

3.1 Conserved process states

For a lumped gas volume i, the principal states are hydrogen mass m_i, internal energy U_i and wall temperature $T_{w,i}$. The governing balances are

$$\frac{dm_i}{dt} = \sum_j \dot{m}_{ji} - \sum_k \dot{m}_{ik}, \tag{1}$$

$$\frac{dU_i}{dt} = \sum_j \dot{m}_{ji} h_{ji} - \sum_k \dot{m}_{ik} h_i + \dot{Q}_i, \tag{2}$$

$$C_{w,i} \frac{dT_{w,i}}{dt} = UA_{gw,i}(T_{g,i} - T_{w,i}) + UA_{wa,i}(T_{\mathrm{amb}} - T_{w,i}) + \dot{Q}_{\mathrm{ext},i}. \tag{3}$$

Pressure and temperature follow from a real-gas property relation,

$$(p_i, T_{g,i}) = \mathcal{P}_{\mathrm{H_2}}\left(m_i/V_i, U_i/m_i\right). \tag{4}$$

The production process calculation uses a versioned property table generated offline from CoolProp; research evaluators also use direct property calls. Pressure is consequently an output of inventory and energy, rather than a prescribed visual trajectory. A leak removes both hydrogen mass and its outflow enthalpy. Closing an inlet cannot, by itself, stop a breach in the remaining vessel inventory.

The Type-IV vehicle vessel uses separate gas, liner and composite-shell energy states. Effective volume and gas-to-liner heat transfer are calibrated globally across the development fills. The structure captures filling heat input and delayed heat exchange without

resolving spatial gas mixing. Where public vessel geometry is unavailable, volume is scaled from 0.122 m^3 per 4.7 kg nominal capacity. The use of an effective geometry is carried into the interpretation of temperature and SOC errors.

Flow restrictions use an effective opening area and discharge coefficient. The research real-gas restriction calculation evaluates the accessible isentropic mass flux,

$$G(p) = \rho(p, s_0)\sqrt{2[h_0 - h(p, s_0)]}, \qquad \dot{m} = C_d A_{\text{eff}} G_*, \tag{5}$$

where G_* is the maximum accessible flux between downstream and upstream pressure. The coefficient and aperture describe a particular boundary; they cannot absorb an unknown pipe network without changing the model's meaning.

3.2 Compression, cascade dispatch and filling control

The compressor uses a quasi-steady positive-displacement representation with suction-density, efficiency and staging assumptions. Interstage cooling and pre-cooling provide the thermal boundary for storage recharge and dispensing. The vehicle controller uses a sampled pressure-ramp request, flow limitation, valve dynamics and protective termination. It represents an APRR-style control boundary, with no proprietary SAE J2601 lookup tables or MC Formula claim.

Cascade selection and recharge are stateful. The selected dispensing bank progresses through the available pressure levels, while recharge uses a lower restart boundary and an upper stop target. This hysteresis avoids instantaneous switching around a single pressure threshold. The high-bank restart band is informed by a disclosed same-site development diagnostic; it is not an approved operating limit. Normal stop targets, relief settings and protective trips are separate inputs so that removing a normal operating target does not silently remove a protection layer.

3.3 Detection and HAZOP evaluation

The versioned HAZOP catalogue connects process nodes to pressure, temperature, flow, gas and flame signals. Each rule retains its condition, persistence, operating-mode qualification and recovery behaviour. The current catalogue has 214 executable simulation rules. This is implementation coverage: it does not establish 214 independently validated hazard detectors.

Mode qualification is essential. Zero flow is expected when a dispenser is idle; it is potentially abnormal only when a transfer is requested and the necessary operating conditions are satisfied. A detector below its threshold can also coexist with a retained alarm. The runtime therefore distinguishes a current observation, an active rule, a latched history and an injected physical fault. Missing or stale input is represented through quality and rule state, rather than being converted to a reassuring numerical zero.

Scenario grouping combines related indications while preserving competing interpretations. A storage gas alarm, pressure loss and an observed release can support one release scenario; a threshold crossing alone does not establish a breach size, ignition or structural failure. A fire scenario has a virtual flame-detection path and separate exposure logic. Detector concentrations are labelled by their calculation origin, including a proxy origin where applicable. They are not treated as spatially validated field measurements.

3.4 Executable actions and completion conditions

The virtual action model distinguishes command issue, valve motion, position feedback and measured flow cessation. Stuck position, internal leakage, partial closure and feedback error can therefore produce a failed action even after a close command has been accepted. The model also provides virtual ESD, bank venting, ventilation, cooling, access restriction and evacuation tracking. The latter are training states; their implementation does not solve pedestrian movement, ventilation CFD or a physical fire-suppression process.

Recovery follows a separate sequence: source repair, leak/functional checks, purge and repressurisation proxies, then restart approval. A new incident invalidates earlier recovery records. A numerical value returning below its alarm threshold does not by itself complete recovery. Likewise, source termination after a confirmed immediate-stop sequence is distinguished from residual-hazard clearance. The current virtual clearance rule requires three consecutive samples with negligible release, less than 0.2 vol% H_2, usable detector quality and no ventilation fault. These values are exercise settings, not facility re-entry criteria.

Every action is followed by process-state evolution and updated release, detector and consequence fields. The framework can schedule paired virtual runs with identical initial states and faults, comparing no action, isolation, ESD and ESD with isolation/ventilation. The feature supplies an experimental instrument for future response comparisons. No reduction in real accident severity is inferred from its availability or from a completed software test.

4 Consequence analysis and SAGA integration

4.1 Release snapshots and consequence endpoints

HyRAM+ 6.1 is connected through a versioned adapter. A calculation request declares the source pressure and temperature, aperture, discharge assumptions, orientation and observation geometry. An active modelled release and a sensor-based hypothetical release are separate request origins. The latter supports conditional analysis but does not establish that a leak exists or that its aperture has been identified.

The outputs distinguish directional 4 vol% hydrogen plume extent from sampled thermal and overpressure criteria. For thermal radiation and unconfined overpressure, the displayed extent is derived from configured observation points exceeding 5 kW m^{-2} or 5 kPa, respectively. It is a sampled screening extent, not an exact continuous contour. A missing crossing remains unestablished; it is not replaced by a default 1 m radius. Visible flame length, plume extent, sampled criterion crossing and the equipment footprint remain different fields throughout the API and visualisation.

Calculation status and time accompany the consequence result. A reused result retains its snapshot provenance; a failed backend call cannot be replaced by a language-model estimate. The dynamic display can compare equipment-relative hazard indices, but its 0–100 score has no calibrated annual frequency or individual-risk interpretation. Three-dimensional overlays are explanatory graphics around equipment, not a numerical mesh of the release field.

4.2 Uncertainty and permitted use

An on-demand sensitivity function perturbs declared pressure, temperature, aperture and discharge-coefficient bounds with a reproducible stratified ensemble. The default bounds are ±3%, ±3 K, ±10% and ±12%, respectively, with 24 samples and seed 731; the implementation caps the ensemble at 128. It reports P05/P50/P95 of available consequence fields and input–output correlations. These are percentiles of the assumed sensitivity ensemble. They are neither experimental confidence intervals nor a calibrated joint probability distribution of station failures.

Each result also carries model identity, validation scope, input quality, uncertainty status and permitted uses. Current component evidence permits screening advice with usable inputs while blocking autonomous actuation based on the consequence result. Deterministic protective logic operates through its own path; an LLM explanation cannot override it. This makes the validation boundary a runtime property of the result rather than a warning confined to the paper.

4.3 Structured response plans

Response plans use five stages: situation confirmation, immediate action, stabilisation, checks before restart and prevention. A step identifies its target equipment, virtual command where available, supporting conditions and completion evidence. Simultaneous compatible indications are consolidated into a response sequence so that the operator does not receive a separate full emergency procedure for every threshold. Conflicting evidence and unavailable inputs remain explicit.

For normal operation, the interface presents monitoring and preventive controls. For abnormal operation it adds the relevant emergency sequence and calculated consequences. A plan is retained as structured deterministic data outside the generated answer. This allows a focused conversational response without discarding the full procedure, and permits a step to be linked to a simulation command whose outcome can subsequently be inspected.

4.4 Direct assistance and document-grounded knowledge

SAGA-PY provides standalone document retrieval as well as the two dedicated digital-twin interfaces. Its document service uses SQLite lexical search and a local hash n-gram index, followed by evidence selection and citation checks. The licensed local corpus is not distributed with the public code. A general document answer and a current-state analysis therefore have different evidence contracts and should not be pooled as one validated capability.

The twin uses a direct-answer configuration. The current sensor or user question is prioritised together with quality, active and retained alarms, release evidence and available consequence results. A compact evidence projection is sent to the provider; the full manifest remains in the local API with a digest linking the two. Raw confidential traces and facility identifiers are not provider context. The main assistant is question-focused, whereas the sensor assistant receives the selected sensor and related process evidence. Periodic analysis is a distinct request type and display path.

The generated answer is checked for unsupported value–unit claims and claims that exceed the registered validation scope. A numerical guard is a limited syntactic/grounding safeguard: it cannot establish that a qualitatively worded recommendation is correct.

Streaming improves presentation responsiveness but does not change the underlying physical result or the evidential status of the answer. Model identity and configuration must be recorded when comparing runs.

5 Data and evaluation methods

5.1 Public filling data and station-side records

Five checksum-verified H2Protocol/Powertech archives supplied 36 Tables Method fills [9]. A pre-model mass-closure screen accepted an integrated measured-flow to SOC-implied inventory ratio of 0.8–1.2. One case was excluded at 1.428. Every third laboratory test formed the 12-case validation set; 23 admissible cases were used for fitting two global multipliers. The experimental time base was retained without fitted shifts or dynamic time warping. A disclosed parser amendment removed an isolated startup pulse from development case L29; the tank fit and all quoted tank results use the corrected normalisation.

For each case, pressure, gas temperature and density-based SOC were scored separately using

$$\mathrm{RMSE}(y) = \sqrt{\frac{1}{n}\sum_{k=1}^{n}[y_k^{\mathrm{predicted}} - y_k^{\mathrm{measured}}]^2}. \tag{6}$$

Case means were bootstrapped with 10,000 replicates and seed 2601. Samples within a fill were not treated as independent experimental replicates.

Confidential operational traces were assessed offline using calibration and disjoint time-ordered holdout partitions. Public outputs contain de-identified counts, relative-time results and approved aggregate errors. They exclude operator, facility, location, manufacturer, exact dates, original tags and raw rows. This is same-site temporal transfer, with no independent-site claim. Pressure-boundary replay, medium/high cascade order and short-horizon recharge pressure forecasting were evaluated separately. A recharge-dwell rule and lifecycle-counter alignment provide retained negative controls on broader interpretation.

The forecast is an empirical adjunct, not a calibration of the conserved station model. Each trace was split 70/30 chronologically before extracting non-overlapping 40 s windows. An observed 10 s loaded-compressor prefix selects the responding bank; a bank-specific robust gain predicts the following 30 s pressure increment. Target selection uses prefix data only. The earlier partition fits the median ratio of future to prefix pressure rates, clipped to $[-1, 3]$. Persistence and uncalibrated prefix-rate extrapolation are comparators. Eligibility requires continuous loaded operation, gaps no longer than 2.5 s and a prefix rise of at least 0.10 MPa with a 1.5 dominance ratio. The primary screens require median/P90 error no greater than 0.5/1.5 MPa, at least 20% MAE improvement over persistence, at least 90% direction agreement and no more than 50% gain shift. This conditional forecast does not predict an unknown compressor-state transition.

A supplementary public intake contained 15 MetHyTrucks workbooks and 58,440 synchronised samples [10, 11, 12]. Metadata limitations, sampling-system rather than vehicle boundaries and prior outcome access restrict these records to descriptive closure and development diagnostics. They are not added to the frozen Type-IV validation population.

5.2 Release, consequence and software verification

The ambient PRESLHY E3.1 evaluation fixed model and protocol hashes before numerical workbook access [13]. Its 22 eligible cases span 0.5, 1, 2 and 4 mm apertures in a 2.815 L vessel. The frozen model was adiabatic, with instantaneous opening and $C_d = 0.8$. A case required pressure NRMSE $\leq 10\%$ of initial absolute pressure and half-gauge-pressure-time error $\leq 20\%$; the aggregate pass threshold was 70%. Numerical failures were retained. Wall heat transfer was subsequently developed on those consumed records and tested on E5.1 [14]. Ambient Part-B used its own separately frozen depletion configuration [15].

Transfer was also examined with INERIS/CEA high-pressure releases, Sandia/SRI mass-flow and pressure-decay traces, and the small-inventory Ekoto release [16, 17, 18, 19]. Digitised curves retained their extraction provenance and declared uncertainty where available. Separate protocols fixed the relevant flow, pressure and half-time screens before the respective comparisons; no universal threshold was imposed retrospectively across physically different experiments.

For the immediately ignited vented-enclosure component, four previously viewed cases were assigned to development and 27 untouched cases to a frozen holdout [20]. A species/energy balance used measured mass-flow histories and common reported coefficients without case-specific fitting. The primary decision required at least 20 cases and at least 80% with peak-pressure absolute error ≤ 2 kPa. A separate 44-case delayed-ignition rank screen tested whether pressure/nozzle inputs preserved severity ordering after instrumentation and geometry-related stratification [21].

HyRAM+ verification compared installed source hashes with public v6.1 sources, ran 47 upstream physics tests and compared three production-adapter cases against direct public-API calls. Translation tolerance was 10^{-12} relative and 10^{-9} absolute. These are package and interface checks, distinguished from independent experimental validation of the station configuration.

5.3 Incident mapping and SAGA comparison

The public HRS subset of HIAD 2.2 supplied 34 historical incidents, 33 with an emergency-action, lesson or corrective field [22]. Scenario mapping and virtual command coverage were evaluated separately from generated-answer quality. Public KHK reports supplied an additional accident-provenance check [23]; incident prose did not set physical model parameters.

The retrospective language comparison used all 34 events. Historical emergency-action, lesson and corrective fields were withheld from model-visible context. One direct SAGA answer per event was compared with deterministic alarm-only text under the archived English, 900-token configuration. The proxy was fixed before cohort answer collection, but the category taxonomy and references had already been used in development. It is consequently a development benchmark rather than an independent prospective hold-out. The archived run records Groq and model identifier `qwen/qwen3.8-27b`; each event contributes one generated answer.

For event e, the proxy is

$$S_e = \tfrac{1}{2}(R_e + C_e), \tag{7}$$

where R_e is expected coarse action-category recall and C_e is coverage of the five response stages. The eight categories span isolation, detection, fire/cooling, evacuation, communication, inspection/repair, procedural controls and ventilation/purge. A paired event-level

bootstrap used 10,000 replicates and seed 20261008. A subsequent audit evaluated reference precision/F1, additional categories and word count on saved guarded answers. That audit is explicitly post-outcome and used a different output revision.

No operator trial or expert correctness rating was conducted. The comparison tests structure and coarse lexical alignment; it cannot isolate the effect of the LLM from the richer context and response templates supplied to it. Historical reports are also incomplete references, so an additional category is not automatically an unsafe recommendation.

6 Results

6.1 Vehicle-boundary thermodynamics

The fitted effective-volume and gas-to-liner multipliers were 1.05273 and 31.46066. Table 2 gives case-mean errors and bootstrap intervals. The 12 validation fills produced mean RMSE of 3.841 MPa, 4.833 °C and 3.850 SOC percentage points. Figure 2 shows the joint pressure/temperature comparison; Fig. 3 exposes the between-fill variation hidden by the means.

Table 2: Type-IV model errors; parentheses are case-bootstrap 95% intervals for the mean.

Set	Fills	Pressure RMSE (MPa)	Temperature RMSE (°C)	SOC RMSE (points)
Calibration	23	4.529 (3.237–6.126)	7.889 (6.072–9.870)	4.455 (3.517–5.455)
Validation	12	3.841 (2.507–5.835)	4.833 (3.707–6.113)	3.850 (2.921–4.909)

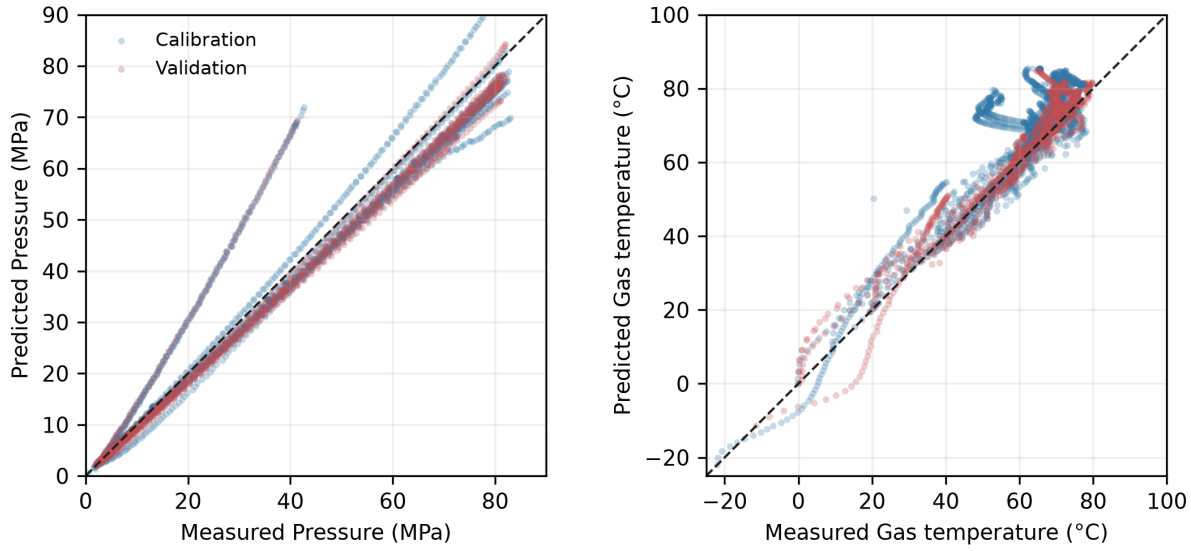


Figure 2: Measured-boundary Type-IV pressure and gas-temperature predictions on the experimental clocks. The dashed line is equality.

Lower validation than calibration means do not establish universal generalisability. The large heat-transfer multiplier is an effective correction, and first-order optimality of 3115.9 despite successful optimiser termination indicates incomplete parameter identification. A prospective Type-III transfer passed pressure at 0.943 MPa RMSE but failed

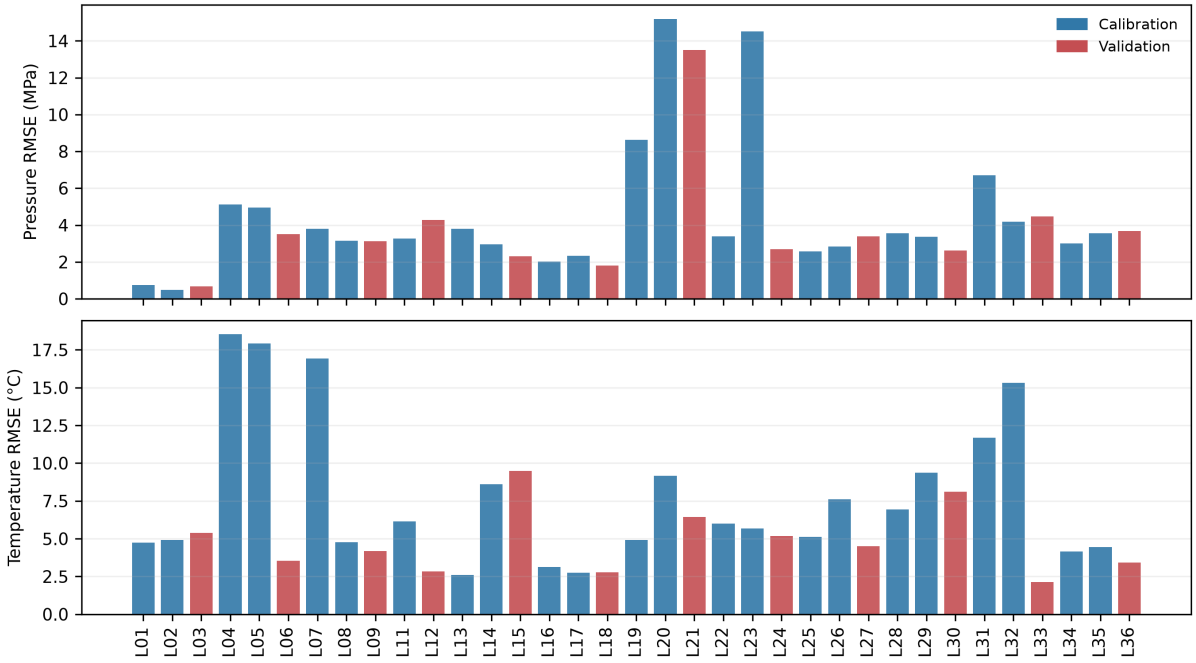


Figure 3: Fill-level errors with frozen validation cases separated from calibration. The pre-model mass-closure exclusion has no bar.

temperature at 19.179 K RMSE. The underlying experiment has a distinct vessel and heat-transfer boundary [24, 25]. Together, these findings support the tested Type-IV boundary while exposing a thermal-transfer limitation.

The supplementary MetHyTrucks audit detected 18 sessions. Nine had changing scale mass and eight of those nine met the descriptive 0.8–1.2 flow–mass closure screen (median ratio 1.003). Unchanged-scale sessions were unscored. Neither this closure result nor a successful sampling-system replay validates vehicle-filling control.

6.2 Station-side holdouts and full-loop transfer

The restricted-data assessment supports several specific station-side results (Table 3). The 394-case recharge-pressure holdout gave 0.184 MPa MAE and 0.529 MPa P90 absolute error; medium/high cascade pairs had a 94.38% sequential fraction among evaluated pairs. Coverage and event definitions are part of these results: the cascade fraction does not represent all station events, and a short-horizon pressure forecast is not a compressor capacity or complete charging-cycle prediction.

Persistence MAE was 0.656 MPa and uncalibrated prefix extrapolation MAE was 0.196 MPa. The calibrated surrogate improved MAE by 71.9% against persistence but only 5.8% against prefix extrapolation. Most of the benefit therefore comes from using the observed pressure trend rather than the fitted gain. Medium/high holdout MAE was 0.116/0.227 MPa across 151/243 cases; the overall positive-direction fraction was 94.9%. These results remain conditional on the eligible loaded-compressor windows.

The dwell rule was disabled after its failed temporal transfer. A separate post-outcome analysis of 124 completed stop/restart cycles gave a median high-stage pressure drop of 4.555 MPa. The rounded 4.5 MPa restart band is therefore an evidence-informed development setting, with no untouched holdout claim. The pressure forecast and cascade checks remain separate from that setting and from the failed lifecycle-counter interpretation.

Table 3: Confidential same-site temporal holdouts. No raw or identifying records are disclosed.

Endpoint	Holdout population	Retained result
Pressure-boundary replay	30 points, 1,755 s	Replay completed; bounded boundary evidence
Medium/high cascade order	3,664 pairs	94.38% sequential; pair coverage 70.39%
Recharge-pressure forecast	394 cases	MAE 0.184 MPa; P90 error 0.529 MPa
Lifecycle-counter alignment	Counter/pressure holdout	Recall 19.60%; pressure precision 90.27%; negative
Recharge time dwell	32 completed restarts	Holdout minimum 176.0 s below fitted 290.0 s; negative

Complete station-to-vehicle transfer was substantially harder. An earlier prospectively frozen MC Default controller configuration passed 0/8 joint screens, with mean pressure, temperature and SOC RMSE of 15.862 MPa, 13.230 °C and 18.082 points. These experiments supplied a pressure–time schedule, whereas the frozen implementation accepted an endpoint-equivalent constant ramp. The result therefore tests that disclosed approximation and historical configuration, not a faithful MC Formula or the final implementation.

Subsequent development corrected parsing, nominal-pressure handling, anti-windup, cooling-class protection and cascade dispatch. Corrected development and already-inspected comparison sets still passed only 1/8 and 2/11 joint screens. Their mean errors were 6.602/4.530 MPa, 11.428/7.819 °C and 7.297/4.625 SOC points, respectively. The consumed prospective set was not used to issue a new positive final-system decision. Thus a successful measured-inlet tank model coexists with unresolved controller and boundary transfer errors.

6.3 Release and consequence applicability

The independent component results are mixed (Table 4). The frozen ambient E3.1 source model passed 11/22 joint screens, below its 70% criterion; six high-pressure cases retained thermophysical-domain failures. Adding wall thermal inertia and real-gas discharge removed those failures and passed 20/22 consumed cases, but this developmental improvement did not transfer to an eligible, sufficiently large E5.1 holdout. Only three cases met its primary eligibility definition and two passed; the protocol required at least four. Ambient Part-B passed 3/5, also below 70%.

The Ekoto transfer passed all frozen mass-flow screens, including 12.79% median absolute percentage error and 6.07% half-peak-time error. Its short, small-inventory release and independently specified coefficient distinguish it from the longer or higher-pressure failed transfers. It therefore provides a bounded positive result, rather than a correction factor for every apparatus.

For the immediately ignited enclosure, all 27 cases met the primary 2 kPa peak-pressure screen. Secondary timing and trace-NRMSE screens passed 21/27 and 26/27. The measured source-flow boundary is important: this validates the enclosure pressure-peak calculation under the tested geometry, not source depletion upstream of that boundary. The unignited enclosure passed 7/10, below its distinct frozen 80% threshold [26].

Table 4: Release/consequence endpoints retained by experiment. Joint decisions use their original protocols; they are not pooled as a single accuracy.

Experiment and endpoint	Joint result	Main interpretation
PRESLHY E3.1 ambient depletion	11/22; failed	Adiabatic source insufficient across tested groups
E5.1 revised-source transfer	2/3 eligible; insufficient	Minimum sample and pass criterion unmet
Ambient Part-B depletion	3/5; failed	Timing errors despite passing pressure screens
INERIS/CEA near 90 MPa flow	0/3 aperture series	Diameter/apparatus transfer unsupported
Sandia/SRI 2006 flow	Joint screen failed	NRMSE 5.83%; median and half-time errors failed
Sandia/SRI 2007 pressure	Joint screen failed	NRMSE 11.58%; half-time error 6.51% passed
Ekoto small-inventory flow	All three screens passed	NRMSE 3.84%; bounded 13.45 MPa, 3.63 L source
Unignited enclosure peak	7/10; failed	Below frozen 80% aggregate criterion
Immediately ignited enclosure peak	27/27 passed	MAE 0.672 kPa; maximum error 1.796 kPa
Large-scale delayed-ignition rank	Controlled screens failed	Pooled association did not survive stratification

The delayed-ignition archive further demonstrates why geometry matters. Across 44 cases, pooled Spearman correlation was 0.572. After stratification, correlation was -0.023, concordance 0.421 and top-third recall 0.294, below their 0.45/0.65/0.50 criteria. Predictions spanned only 22.091–22.740 kPa against measured maxima of 0.220–235.390 kPa. Source pressure and aperture alone did not preserve controlled severity ordering; this result is not converted into a station explosion-distance claim.

6.4 Software translation and executable response coverage

The installed HyRAM+ source matched the public v6.1 source over 38 Python files. All 47 selected upstream tests, comprising 803 subtests, passed; all three production-adapter parity cases met the specified numerical tolerance. The low-bank 1 mm, cold mid-bank 2 mm and high-bank 3 mm cases returned directional 4 vol% plume extents of 4.339, 11.382 and 17.224 m. These are example model outputs under declared inputs, not measured site boundaries.

All 34 HIAD cases mapped to registered response families. Replay covered 29 through direct process representations, four through a compressor proxy and one structural-damage case through response logic only. The five active-response categories produced 72 event–category obligations with virtual commands. Ten family templates executed 52 commands with completion feedback. The counts establish mapping and execution coverage, not independent effectiveness of every action. KHK citation-based metadata provided a separate check across 23 reports and 26 incident codes; eight registered response recipes passed their integration checks.

An archived full Python regression run reported 1,650 passed, no failures and no

skipped tests, with 23 warnings. It is an implementation snapshot at the recorded code revision, not a physical-accuracy percentage. Current extensions also provide partial-closure, stale/dropout and residual-hazard checks, plus paired-run and sensitivity interfaces. Their implementation status is kept separate from the earlier empirical evaluations and does not enlarge those experimental samples.

6.5 SAGA response structure and information burden

The archived 34-event comparison favoured SAGA on its declared machine proxy (Table 5). Mean category recall was 0.524 versus 0.189 and stage coverage 0.982 versus 0.600. The mean proxy difference was 0.360 with a paired bootstrap interval of 0.309–0.411. The first run flagged unsupported value–unit claims in 3/34 SAGA answers; a later guarded rerun reduced the flagged count to 0/34. That rerun is a post-outcome engineering correction, not the original benchmark result or evidence that all unsafe statements were removed.

Table 5: Original retrospective HIAD comparison, one answer per event.

Metric	Alarm-only	SAGA
Events	34	34
Mean expected-category recall	0.189	0.524
Mean five-stage coverage	0.600	0.982
Mean response-structure proxy	0.401	0.760
Answers with unsupported numeric flags	0	3
Failed calls	0	0

The saved guarded-output audit on 33 reference-evaluable events exposes a trade-off. Reference F1 was 0.564 versus 0.281, while precision was 0.680 versus 0.697. Mean word count rose from 28.1 to 140.5 and mean non-reference category count from 0.303 to 0.970. Greater coverage therefore came with more information to process. The display design uses a focused answer with the deterministic full plan separately available; this is a design response to the burden, not a demonstrated reduction in operator workload.

7 Discussion

7.1 What the integrated workflow adds

The principal result is the connection between a safety statement and its checkable process context. A gas alarm can be displayed together with its quality, a possible release source, a declared consequence request, applicable isolation commands and the measurements needed to confirm success. The next simulation sample then updates the process instead of merely marking the procedure complete. This gives the demonstrator a causal training structure even where the quantitative physical fidelity remains limited.

Consider a storage-vessel leak. Isolating connected inlet/outlet paths changes the connected inventory, but the remaining vessel can continue to release. Ending the source input after an exercise stop sequence still leaves a separate residual-hazard state. A requested valve closure can fail its feedback or flow condition. Keeping these distinctions visible prevents a textual answer, accepted command or cleared badge from serving as

its own evidence of safety. This is the rationale for combining physical state, HAZOP, consequence screening and staged guidance within one traceable workflow.

The contribution is not a new universal release correlation or an independently validated autonomous safety policy. It is an integration method that preserves the evidence boundary of its components. The direct SAGA interface adds question-specific explanation to this structure; the standalone retrieval service supplies a separate document-knowledge capability whose citation quality requires its own evaluation.

7.2 Interpreting the physical evidence

The strongest filling result is the measured-boundary Type-IV model. Supplying the experimental inlet flow and temperature removes controller and upstream uncertainty from that test. It is therefore compatible with a negative station-to-vehicle result: an adequate downstream model can be driven by an inadequate upstream schedule. The large effective heat-transfer correction and failed Type-III temperature transfer warn against interpreting an aggregate temperature error as an identified material parameter.

The confidential data add a different form of evidence. A useful same-site recharge-pressure forecast and a largely sequential cascade pattern help ground station-side behaviour, but do not provide missing vehicle inlet, controller or receptacle measurements. The failed dwell and counter checks show that temporal event interpretation cannot simply be inferred from a plausible pressure trajectory. These records improve the realism of bounded operational assumptions without making the public component and confidential station samples interchangeable.

Release results are similarly boundary-dependent. Vessel-wall heat and line/manifold inventory matter for depletion, while the successful enclosure pressure model is driven by measured source flow. HyRAM+ adapter parity removes one integration uncertainty, but cannot repair an incorrect release boundary. For this reason the interface preserves both the point calculation and the input/domain flags. Failure to reproduce geometry-controlled explosion ranking also precludes converting a visually convincing overlay into a validated equipment hazard zone.

7.3 Meaning and limits of the SAGA evaluation

The retrospective comparison demonstrates that the linked configuration produced more of the selected action categories and response stages than the compact alarm baseline. It does not demonstrate that a language model selected the safest action. Richer context, deterministic templates and longer output are all plausible contributors; the study contains no factorial ablation that separates them. A single answer per event also leaves provider/model stochasticity unmeasured.

The unsupported-numeric findings motivate a guard, while the precision and length findings motivate prioritisation. Neither supplies a complete safety assurance argument. A clinically or industrially consequential omission can receive a high lexical score, and a prudent action absent from a short historical report can lower measured precision. Expert assessment of action correctness, facility-specific review and operator testing are therefore prerequisites for operational deployment, outside the endpoints measured here. KGS-collected, de-identified research material and AI-assisted organisation informed the preliminary evidence and response resources; they do not replace that review.

7.4 Limitations and reproducibility

All instruments, interventions and scene effects are simulated. Geometry, weather, occupancy, heat-transfer coefficients, compressor performance and most actuator failure parameters remain scenario assumptions. Spatial detector response and ventilation are not independently validated at station scale. Detailed structural failure, realistic evacuation timing, spray/fire interaction and full probabilistic QRA are beyond the demonstrated physical scope.

No openly reusable synchronized station-to-vehicle raw cohort with the required controller, thermal and vehicle channels was identified. The full-loop result therefore remains a disclosed negative transfer diagnostic rather than a primary final-system accuracy claim. Sources containing plots, aggregate operating summaries or inaccessible raw files were not promoted to substitute holdouts. Future validation should prioritise a defined apparatus and endpoint with complete units, geometry and common-time measurements, rather than an ever-larger collection of loosely compatible records.

Public scripts, hashes and result manifests permit inspection of the reported comparisons, while restricted raw records remain owner-controlled. Readers can reproduce the public numerical experiments and software interfaces subject to third-party access and licensing. They cannot independently reconstruct private station traces from the published aggregates. The release-network candidate, paired intervention interface and bounded sensitivity function are available for future tests; no unreported empirical success is inferred from them.

8 Conclusions

This study developed a simulation-based hydrogen-station safety digital twin that links process dynamics, HAZOP deviations, consequence calculations and SAGA guidance to executable virtual responses. Its defining feature is the preservation of distinctions between a signal, an inferred scenario, a calculated consequence, an action command and confirmed simulated completion.

Evaluation supports several bounded results: measured-boundary Type-IV prediction at 3.841 MPa pressure and 4.833 °C temperature RMSE; same-site recharge-pressure forecasting at 0.184 MPa MAE; a 27-case ignited-enclosure peak-pressure result at 0.672 kPa MAE; and verified HyRAM+ translation. Negative controller, depletion and geometry-transfer results define where those findings do not apply.

SAGA increased retrospective response-structure coverage, with a proxy of 0.760 against 0.401 for alarm-only text, while a separate audit exposed longer answers and additional action burden. The combined outcome is a reproducible research and training framework with traceable numerical and procedural boundaries. Field deployment requires the facility-specific measurement, HAZOP, expert and operational evaluations appropriate to the intended use.

Data and code availability

The digital-twin source, evaluation scripts, result manifests and documentation are available at `https://github.com/lyullee/hydrogen-station-sim`, archived as 10.5281/zenodo.23084421. SAGA-PY is available at `https://github.com/lyullee/saga-py`, archived as 10.5281/zenodo.23084441. Project code is MIT licensed; third-party

data and dependencies retain their own terms, including GPL-3.0 for HyRAM+. The licensed document corpus, local indexes and provider credentials are not included.

Confidential station records are not deposited. Only approved de-identified aggregates, relative-time results and validation manifests are disclosed. Company, site, location, calendar dates, manufacturer, equipment identifiers, original tags and raw records remain in the owner-controlled environment. The public record supports inspection of the reported methods and aggregates, not independent reconstruction of the restricted measurements.

CRediT authorship contribution statement

Ugwiyeon Lee: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Validation, Visualization, Writing – original draft, Writing – review and editing. Seong-Yun Bae: Writing – review and editing. Hakrae Kim: Writing – review and editing. Jungsu Lee: Writing – review and editing. Jinseo Kim: Writing – review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used ChatGPT and Codex (OpenAI) to assist literature searches, manuscript drafting and language revision, and the development of analysis and plotting code. Figures were rendered from archived numerical data or explicitly stated screening criteria using reproducible Python code, not generative image synthesis. The authors are responsible for reviewing the cited sources, code, numerical results and manuscript, and take full responsibility for the published article.

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